INCH-POUND

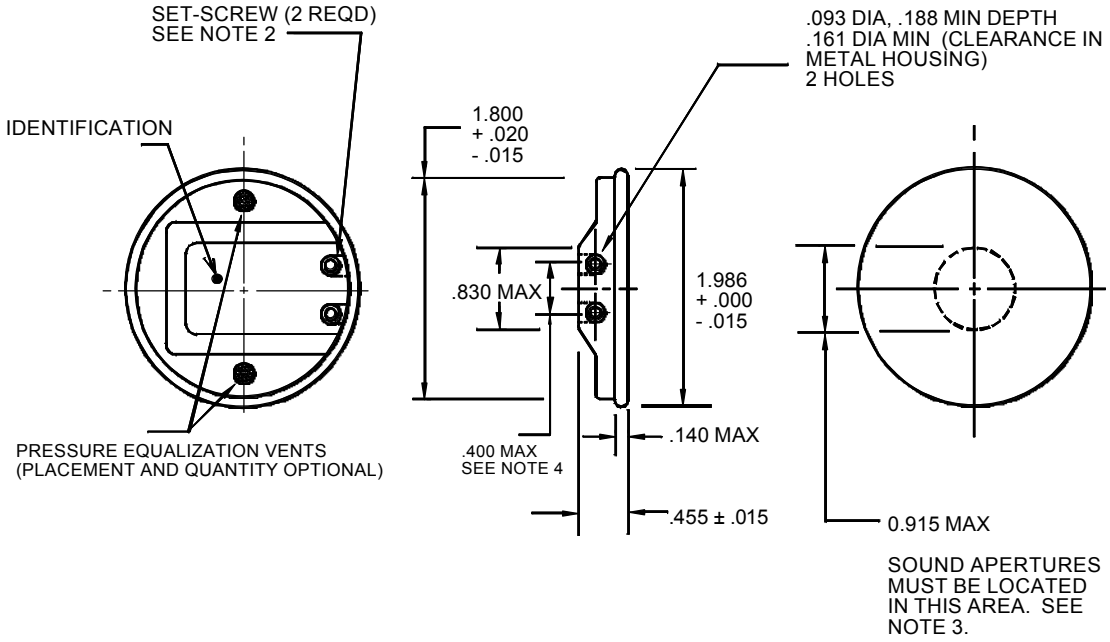
MIL-PRF-25670/4(USAF)
30 May 1997

PERFORMANCE SPECIFICATION SHEET

EARPHONE ELEMENT,
HIGH- AND LOW-ALTITUDE,
M25670/4-01

This specification sheet is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-25670.



NOTES:

1. Dimensions are in inches. Unless otherwise noted, tolerance is ± .010 in.
2. Screws shall be recessed, of type slotted or allen (for interchangeability of spares), and shall securely hold the next-higher-assembly headset connector-pins, preventing inadvertent disconnection.
3. All outer dimensions shall ensure a firm positioning of the element in standard earcup foam filler material (depth, contour, ear-position), as well as within the standard earcup retaining-ring.
4. Cable-connection holes shall be positioned within reach of standard-length headset cabling.
5. Back-cover dimensions or contours without assigned numerical measurements are optional.

FIGURE 1. Earphone element M25670/4-01.

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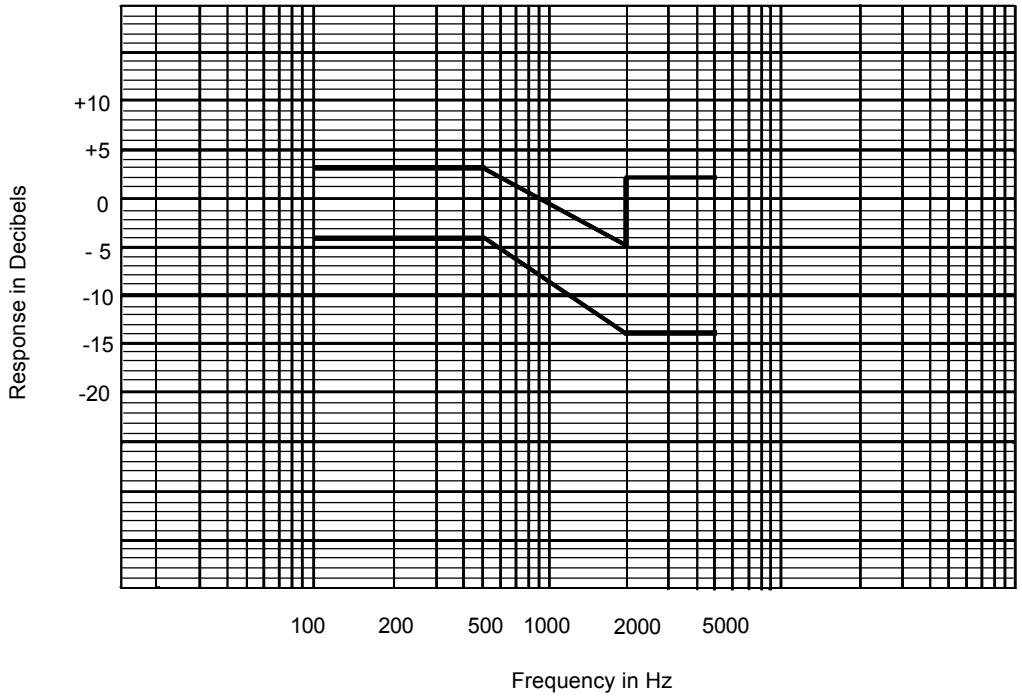
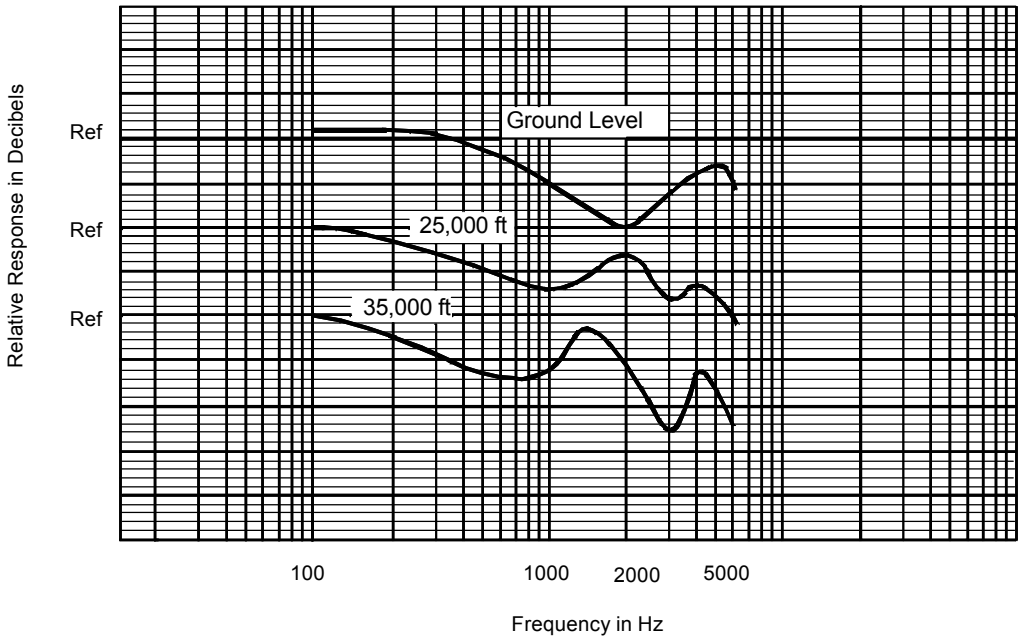


FIGURE 2. Ground level frequency response envelope for earphone element M25670/4-01.



NOTE: Curves are plotted with 10 dB vertical spacing for clarity of presentation.

FIGURE 3. Typical pressure frequency response characteristics for earphone element M25670/4-01.

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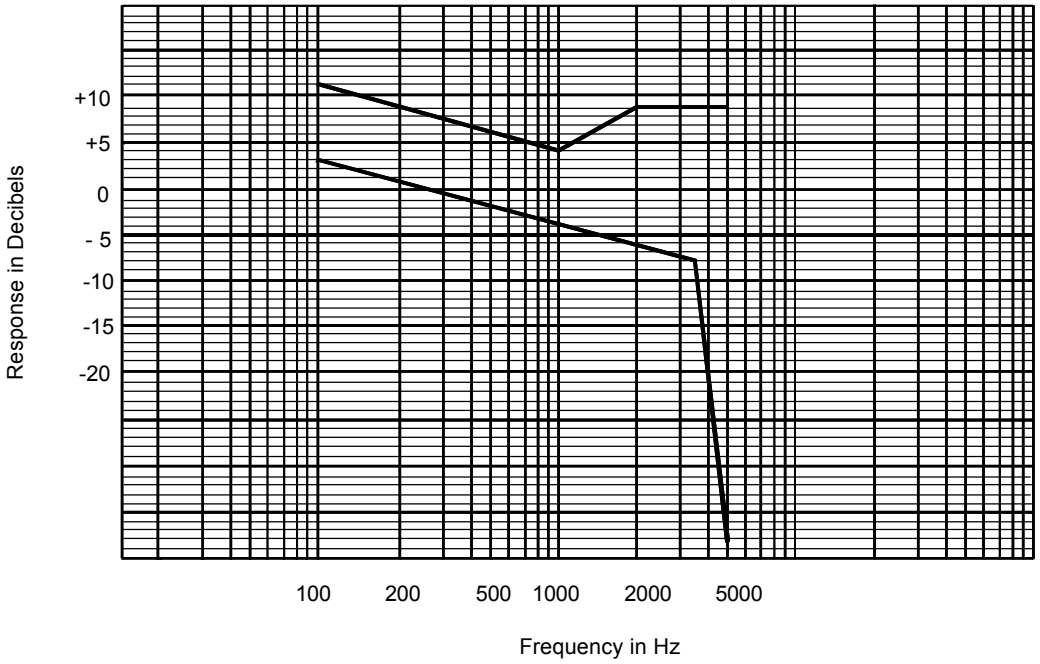


FIGURE 4. 25,000 ft frequency response envelope for earphone element M25670/4-01.

REQUIREMENTS:

Dimensions and configuration:	See figure 1.
Maximum weight:	22 grams.
Color:	The back case shall be dull black, or an equivalent low-contrast, dark color, while the front cover shall be clear, anodized aluminum or black, in order to indicate the versatility of the part's use in both ground-level and in altitude applications.
Stray magnetic field:	5° maximum at 12 in.
Frequency response range:	200 to 5000 Hz.
Frequency response:	
At ground level :	Within the limits specified in figure 2.
At altitude:	Within the characteristics shown on figure 3 and in the response envelope in figure 4.
Sensitivity:	105 ± 3 dB at 1000 Hz at ground level; 100 dB minimum at 1000 Hz at an altitude of 25,000 feet.

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Impedance:
at 1000 Hz - 19 ± 2 Ω.
between 100 to 3000 Hz - Shall not exceed 22 Ω.

Harmonic distortion:
between 100 and 2000 Hz - 3%, maximum.
between 2000 and 5000 Hz - 5%, maximum.

Intelligibility: In accordance with MIL-PRF-25670.

Part or Identifying Number (PIN) shall be: M25670/3-01.

Interchangeability: Earphone element M25670/4-01 is interchangeable with earphone element H-143/AIC.

Intended use: Earphone element M25670/4-01 is a low-impedance, light-weight transducer to be used in noise-attenuating headsets, requiring voice communications at both ground level and at altitude. This part is especially suited to headsets used in high-noise applications, as it allows more signal in the upper-frequencies of the human voice range than the other earphone elements associated with MIL-PRF-25670, thereby providing more perceptible offset to the high-frequency noise common to certain transport and fighter aircraft engines.

CONCLUDING MATERIAL

Custodians:
Air Force - 85

Preparing activity:
Air Force - 85

Review activities:
Air Force - 99
DLA - CC

(Project 5965-0254)

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